

MUHAMMAD MUJAHID TUFAIL

AI BACKEND DEVELOPER

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PROFESSIONAL SUMMARY

AI Backend Developer who architects and ships production systems end to end — not prototypes. Builds scalable backend infrastructure (FastAPI, REST APIs, PostgreSQL, MongoDB) with LLM integration, RAG architectures, and multi-agent pipelines using LangGraph, LangChain, and Anthropic, OpenAI, and Azure AI APIs. Comfortable owning the full lifecycle: system design, database architecture, cloud deployment (AWS, Docker), and production-grade reliability.

TECHNICAL SKILLS

Backend & APIs: Python, FastAPI, REST APIs, JWT Auth, SSE Streaming, SQLAlchemy, Pydantic, SlowAPI, Docker, Swagger/OpenAPI, BeautifulSoup

Databases: MongoDB, PostgreSQL, SQLite, FAISS, pgvector, ChromaDB

AI & Generative AI: LangChain, LangGraph, RAG Pipelines, Anthropic API, OpenAI API, GPT-4o, Llama 3, Groq, Gemini, Tavily Search, LangSmith, Prompt Engineering, Azure AI Speech

NLP & ML: SentenceTransformers, Cross-Encoder Reranking, scikit-learn, PyTorch, Keras, spaCy, NLTK

Languages & Tools: SQL, C++, C#, Git, GitHub, Railway, Render, Vercel, AWS EC2, Neon

PROJECTS

Restaurant POS System — Dowdy Pizza (Client) | FastAPI, PostgreSQL, React, Vercel, Render, Neon

Live: <https://dowdy-pos.vercel.app/>

- Built a full-stack POS system for a live restaurant client — FastAPI backend, PostgreSQL database, and AI-scaffolded React frontend covering order management and billing end-to-end.
- Deployed across 3 free-tier services (Vercel + Render + Neon); implemented an automated cron-based keep-alive to eliminate Render free-tier cold-start latency and maintain consistent uptime.
- Delivered to an active paying client, iterating directly on real-world restaurant workflow feedback.

Full Stack AI Chatbot API | FastAPI, PostgreSQL, AsyncAnthropic, Docker

GitHub: <https://github.com/MujahidMalik7/Full-Stack-AI-Chatbot-API>

- Engineered a production-deployed conversational API with JWT auth (bcrypt, 30-min expiry), real-time SSE streaming via AsyncAnthropic, full CRUD conversation management, role-based access control, and IP-based rate limiting (20 req/min via SlowAPI).
- Dockerized with Docker Compose; PostgreSQL in production on Render; implemented sliding window context (last 20 messages), Pydantic input validation, and graceful stream error handling with transactional rollback on failure.

PROFESSIONAL EXPERIENCE

AI Backend Developer (Python) | ARGONTEQ, Lahore

Feb 2026 – Sep 2026

AI Knowledge Assistant — Multi-Tenant SaaS RAG Platform

- Architected a multi-tenant SaaS RAG platform enabling companies to query internal PDF/DOCX knowledge bases via AI chatbot — tenant-level vector isolation in pgvector, 3-tier RBAC, and JWT auth ensuring zero cross-tenant data exposure.
- Engineered a RAG pipeline with hybrid search (keyword + vector retrieval), SSE-streamed responses, and per-user token cost tracking across 25+ endpoints — improving retrieval accuracy and giving admins full AI spend visibility.
- Deployed via Docker Compose; built a React frontend with role-based dashboards, admin analytics, and real-time markdown rendering of streamed AI responses.

WeNet — Game Backend, AI Integration & Server Management

- Closed a resource exploitation vulnerability via an anti-farming system with MongoDB-backed state persistence; migrated storage from raw binary to FileZilla-hosted URLs, cutting payload sizes and improving response speed.
- Rebuilt the player profile system — episode tracking, reward history, rank titles, and last-played state — enabling accurate long-term progression tracking; introduced an API-layer hint-penalty rule enforcing reward fairness.
- Engineered a hybrid AI evaluation pipeline (Azure AI Speech + GPT-4o-mini), cutting evaluation costs ~90% while improving output quality; integrated FPT.ai for KYC identity/age verification supporting child-safety compliance.
- Built a dynamic admin dashboard with dual-approval workflow and immutable audit logs for real-time reward and resource management.
- Deployed and manage AWS EC2 infrastructure; refactored reward distribution into a config-driven architecture and maintained stability across 50+ FastAPI endpoints.

EDUCATION

University of Management and Technology | Lahore, Pakistan

Nov 2021 – Aug 2025

- Bachelor of Science in Computer Science (BSCS) | CGPA: 3.54
- Courses: Artificial Intelligence, Natural Language Processing, Machine Learning, Database and Management, Object Oriented Programming, Data Structures & Algorithms